

Fiscal Amenities and Spatial Inequality: The Regional Impacts of Italy's 2017 Non-Dom Tax Regime

Research Question. How did Italy's 2017 non-dom tax regime affect regional housing prices and housing-burden inequality, and were these effects stronger in high-amenity regions, as predicted by spatial equilibrium models?

(Candidate Numbers)

XPXF3

WHYX3

YCGL6

XPRN1

Word count: 1486

Introduction

Italy introduced a “non-dom” tax regime in 2017, offering eligible High Net-Worth Individuals (HNWIs) to pay an annual flat levy of €100,000 on foreign income, substituting progressive marginal rates reaching 43%, serving as a tax break (KPMG 2017).

Since its introduction, nearly 1,500 individuals have enrolled in it as of June 2023, generating €315 million in tax revenue between 2020-2023 (Reuters 2025). The policy coincides with a broader surge in inward mobility with a record 3,600 millionaire migrants projected in 2025 (Henley & Partners 2025). Whilst multiple factors could contribute to this trend, the non-dom regime represents a major new fiscal amenity shaping Italy’s attractiveness to wealthy foreigners.

This paper examines whether the reform altered Italy’s regional spatial equilibrium by disproportionately attracting high-income migrants toward high-amenity regions, thereby affecting housing prices and housing cost inequality. This draws upon the Rosen-Roback prediction that place specific amenities capitalises into housing markets with heterogenous regional effects.

Literature Review

Our theoretical foundation lies in the Rosen-Roback (1982) model which provides a framework for analysing spatial equilibrium. Subsequent formalisation reinforces how amenities capitalise into higher housing costs and compressed real wages (Glaeser and Gottlieb 2009, 987-989). More recently, Allen & Arkolakis (2023, 19-20), extended this model to incorporate supply and demand linkages, emphasising that place-based policies can be interpreted as shifting the local supply curve via amenity feedback, disrupting spatial equilibrium.

Empirical studies support the Rosen-Roback model's prediction that migration and sorting affect local rent. Cun and Pesaran's (2022, 12-14) extension of the model studies the interaction of migration flows with housing prices, modelling how a demand-side shock for highly skilled workers causes in-migration, resulting in higher rents. Similarly, Diamond (2016, 44) estimates how changes in local demand and endogenous amenities lead to a sorting effect of high-skilled workers towards high-wage, high-rent cities.

Nevertheless, few studies have evaluated the effects of fiscal policies on migration and spatial equilibrium outcomes. This is particularly salient given that Colas and Hutchinson (2021, 118-120/124-127) simulated Rosen-Roback model, which shows that regional tax differences can induce migration and income-based sorting, increasing housing demand and prices in favoured areas.

Internationally, studies found evidence of tax-induced mobility among professional footballers in Europe (Kleven, Landaïs and Saez 2013, 1921-1923). Further expansion shows cross-country evidence that differential taxation induces international migration of top earners, with mobility elasticities ranging between 0.1-1.6 (Kleven et al. 2020, 120-122).

This paper aims to synthesise a comprehensive analysis into the effects of tax policy on international migration and its subsequent outcome within a spatial equilibrium framework—drawing on the Italian policy as a natural experiment. By combining spatial equilibrium theory with tax-incentivised migration, we empirically analyse whether a fiscal amenity for HNWIs generates detectable regional effects on housing prices and affordability. The analysis contributes to two strands of literature: (i) work on amenities and spatial sorting, and (ii) the emerging empirical literature on tax-induced mobility and local general-equilibrium effects.

Data

We use annual Italian regional data from 2011–2024, defining 2011–2016 as the pre-policy period and 2017–2024 as the post-policy period, consistent with the introduction of the non-dom tax reform in 2017. Housing prices (HPI) are sourced from the *Agenzia delle Entrate*, while housing-cost burdens, crime rates, tourism indicators, and demographic variables are from ISTAT.

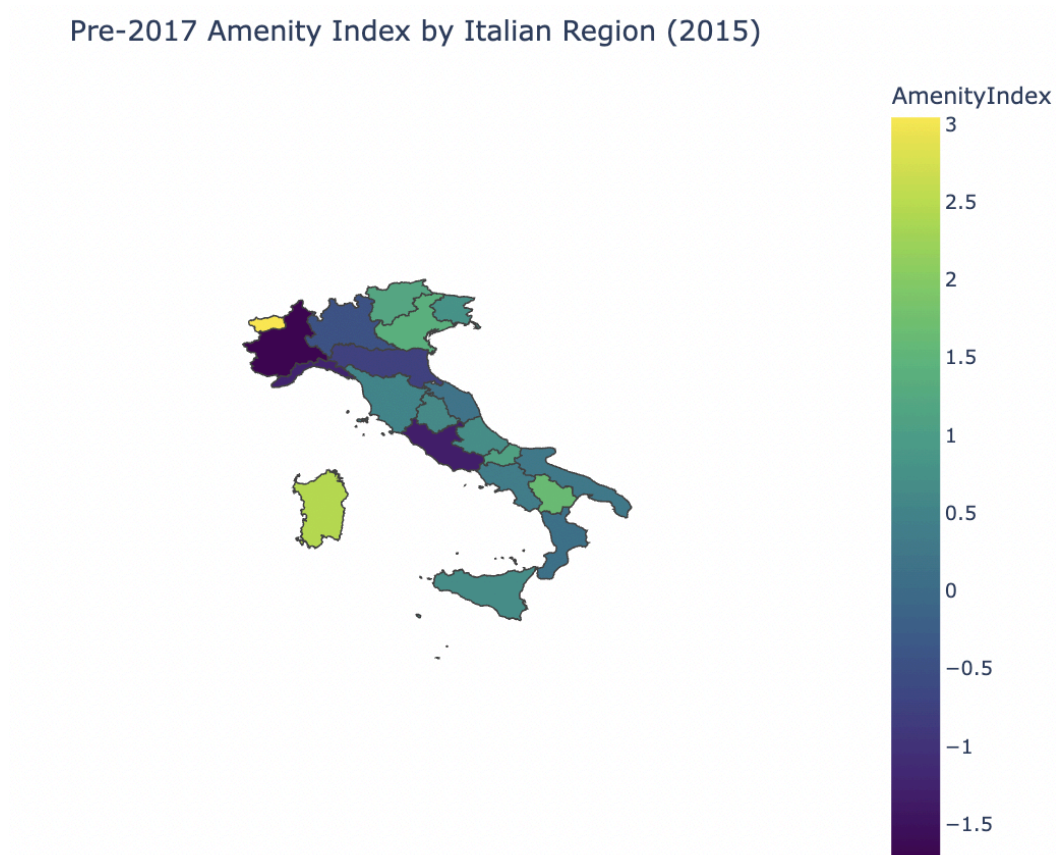


Figure 1. Choropleth Map of the Regional Amenity Index (NUTS-2, Italy)

This map illustrates the spatial distribution of the constructed Amenity Index, combining pre-treatment tourism intensity, crime rates, and coastal access. Higher values correspond to more lifestyle-oriented attractiveness. Some regions score lower due to higher crime z-scores in the pre-period. The index captures non-economic quality-of-life factors rather than productivity..

Average HPI per Italian region per year (slider = year)

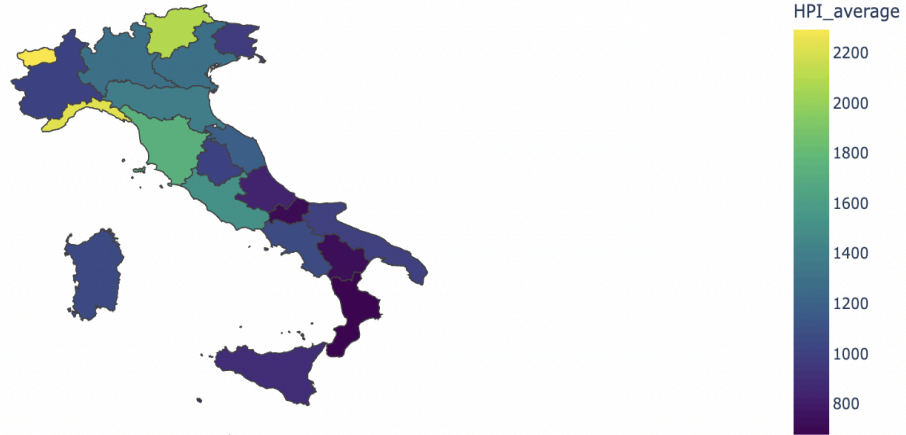


Figure 2. Average Housing Prices Across Italian Regions in 2015, An interactive version throughout the years is available at: [Link For Interactive Figures](#) document name: “Regional Housing Prices Trends, per year 2011–2024.”

This map illustrates large and persistent regional differences in housing prices, with northern regions displaying systematically higher levels relative to the South. This variation motivates our empirical analysis of whether the 2017 non-dom regime shifted regional price trajectories

Amenity Index

Since the non-dom regime was implemented nationwide in 2017, identification relies on cross-sectional variation in treatment intensity, rather than variation in treatment timing. High-amenity regions are presumably more attractive to HNWIs and therefore more affected by the reform.

We construct a time-invariant *Amenity Index* using pre-2017 regional characteristics:

1. Foreign tourist nights per capita,
2. Crime rate per 100,000 inhabitants,
3. A coastal dummy.

Standardising tourism and crime into z-scores, the index is defined as

$$AmenityIndex_r = Z_r^{Tourism} - Z_r^{Crime} + Coastal_r$$

Since region-level counts of non-dom applicants are unavailable, the index serves as a theoretical proxy for heterogeneous exposure to the reform. It captures lifestyle-oriented features rather than economic performance. This aligns with literature on spatial equilibrium, which treats amenities and productivity as separate determinants of regional attractiveness (Roback, 1982).

The main outcomes are:

- Housing prices: log of regional HPI
- Housing-cost burdens: the share of disposable income devoted to housing expenditures.

They capture the channels through which spatial equilibrium predicts adjustments following a demand shock: price capitalisation and affordability pressures.

The introduction of the non-dom regime determines the pre- and post-policy periods used in the empirical analysis.

	log_HPI_average	Observation_Income_Percentage	AmenityIndex	foreign_pc_avg	crime_avg
Mean	6.60	11.67	0.55	2.13	3497.32
SD	1.13	1.27	1.31	2.00	744.19

Table 1. Extracted Summary Statistics

Mean housing prices (log HPI) and housing-cost shares show substantial regional variation, while the amenity index displays meaningful dispersion across regions. Tourism intensity (foreign_pc_avg) and crime levels (crime_avg) also vary widely.

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
Year	14	0	2017.5	4.0	2011.0	2017.5	2024.0
population	280	0	2988691.3	2443246.4	122877.0	1747200.5	10027602.0
area_km2	75	10	15845.2	7090.8	4460.4	16226.7	25835.3
density	253	10	194.6	107.4	52.9	166.3	426.3
gdp	235	16	97219.6	94227.5	6237.0	62605.0	441407.3
gdp_per_capita	235	16	27478.6	7011.5	16442.2	27279.2	44244.7
HPI_average	280	0	1228.9	497.1	639.6	1048.7	2539.8
log_HPI_average	280	0	6.6	1.1	3.3	6.9	7.8
Observation_Disposition_Income	260	7	53929.5	50329.0	2371.9	29202.7	256325.8

Observation_Average_Monthly_Exp	143	0	298.9	47.7	196.0	304.5	400.0
log_Observation_Average_Monthly_Exp	143	0	5.7	0.2	5.3	5.7	6.0
Observation_Income_Percentage	59	0	11.7	1.3	8.4	11.6	15.2
foreign_pc_avg	32	0	2.1	2.0	0.1	1.6	9.5
crime_avg	36	0	3497.3	744.2	2272.9	3388.2	4727.7
coastal_dummy	2	0	0.5	0.5	0.0	1.0	1.0
z_tourism	32	0	-0.0	1.0	-1.0	-0.3	3.7
z_crime	36	0	-0.0	1.0	-1.6	-0.1	1.7
AmenityIndex	41	0	0.6	1.3	-1.7	0.6	5.1
Post	2	0	0.6	0.5	0.0	1.0	1.0

Table 2. Summary Statistics for Regional Panel Dataset (2011-2024)

The variables in this dataset include demographic, economic, and housing-market indicators, and the components used to construct the Amenity Index. It summarises the key outcomes and covariates across Italy's NUTS-2 regions over the sample period.

Empirical Analysis

This section outlines how we estimate the heterogeneous, regional housing-market impact of Italy's non-dom regime. Building on the Rosen-Roback framework, a fiscal amenity that raises demand of a region for HNWIs should increase local housing demand, especially in regions with high pre-existing amenities. This yields two hypotheses:

H1: high-amenity regions experienced larger post-2017 increases in housing prices;

H2: housing-cost burdens rose more in high-amenity regions.

Difference-in-Differences

To test these hypotheses, we estimate a continuous treatment intensity DiD (de Chaisemartin & D'Haultfoeuille, 2022) with region and year FEs:

$$Y_{rt} = \beta(AmenityIndex_r \times Post_t) + \alpha_r + \gamma_t + \epsilon_{rt}$$

where Y_{rt} is either (i) log housing prices or (ii) share of income spent on housing. $Post_t = 1$ for $t \geq 2017$; α_r and γ_t are FEs. Standard errors are clustered at the regional level.

Because $AmenityIndex_r$ is time-invariant and $Post_t$ is uniform across regions, both are absorbed by FEs. Thus β captures whether post-2017 changes are systematically larger in high-amenity regions.

Identification comes from cross-sectional heterogeneity in exposure to amenity shock. A key assumption is a parallel-trends condition in differences: absent the policy, high- and low- amenity regions would have exhibited similar trends in housing prices, conditional on FEs. Figure 2 provides a visual comparison of both groups exhibiting comparable downward movements pre-2017, supporting the plausibility of the assumption.

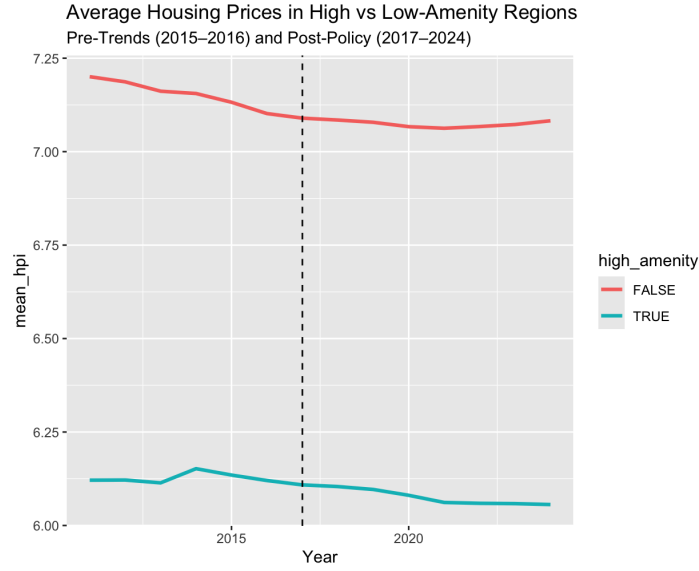


Figure 3. Pre-Policy Housing Price Trends by Amenity Level

High- and low-amenity regions displayed comparable pre-2017 trends, supporting the DiD identifying assumption. Post-2017 patterns show limited divergence, consistent with a small treatment effect.

Continuous Event Study

Since treatment timing is uniform, a staggered-adoption event-study design is not applicable. Instead, a continuous event-study is implemented, interacting the Amenity Index with relative-year indicators

$$Y_{rt} = \sum_{k \neq -1} \delta_k (\text{AmenityIndex}_r \times 1\{\text{RelYear}_t = k\}) + a_r + \gamma_t + \epsilon_{rt},$$

where $\text{RelYear}_t = t - 2017$ and $k = -1$ is the omitted reference year.

This serves as a diagnostic tool, providing visual evidence on whether pre-2017 differential trends exist and whether any post-2017 pattern is consistent with the average treatment effect captured in the DiD model.

Results

	HPI Regression	Inequality Regression
AmenityIndex	−0.009 (0.042)	−1.211** (0.403)
AmenityIndex × Post	0.023 (0.013)	0.198* (0.093)
Num.Obs.	280	280
R2	0.998	0.800
R2 Adj.	0.998	0.772
RMSE	0.04	0.57
Std.Errors	by: Region	by: Region
FE: Region	X	X
FE: Year	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3. Difference-in-Differences Estimates of the Effect of Regional Amenities on Housing Prices and Housing-Cost Burdens.

Estimates from regressions of log housing prices and housing-cost burdens on the interaction between the regional Amenity Index and a post-2017 indicator. All models include region and year fixed effects with standard errors clustered by region. The housing-price effect is positive and marginally statistically significant, while inequality effects remain imprecisely estimated.

Housing prices

Table 3 reports the DiD estimate for $\log(HPI_{rt})$. The coefficient on $AmenityIndex \times Post$ is

$$\hat{\beta} = 0.0225 \text{ (s.e. } 0.0131, p = 0.102\text{)}.$$

A one-standard-deviation increase in the amenity index is associated with a ~2% post-2017 increase in housing prices relative to low-amenity regions. The effect is positive but only marginally significant, consistent with the limited scale of non-dom uptake and the depth of regional housing markets.

Percentage change in regional housing prices (post – pre period)

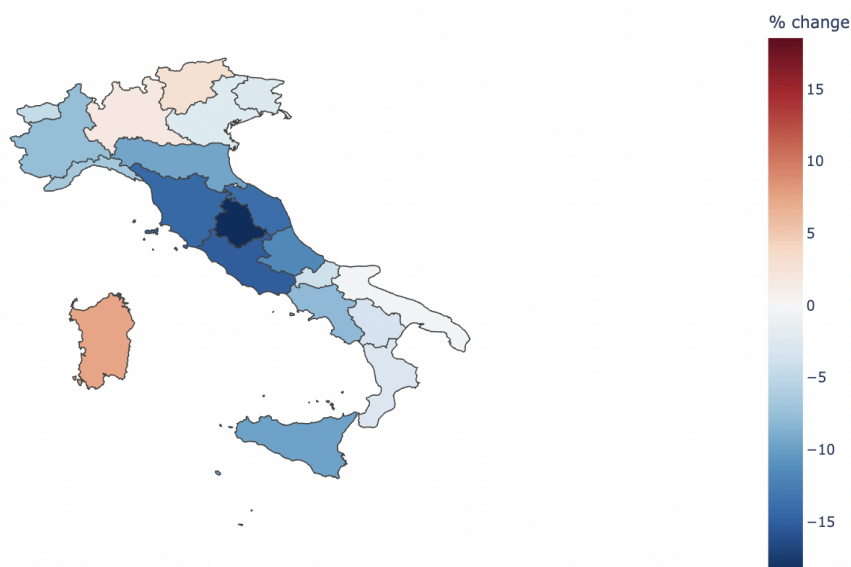


Figure 4. Average Housing Price Change: Post-Reform (2017–2024) vs Pre-Reform (2011–2017)

Percentage change in regional housing prices between the pre-period (average House Prices per region 2011–2016) and the post-period (average House Prices per region 2017–2024). For each region, we compute the mean HPI over the pre-period and over the post-period, and express the difference as a percentage of the pre-period mean. The map displays a variance between High-amenity regions: eg. Trentino-Alto Adige, Lombardy and Sardegna, vs Low-amenity regions (south Italy)

The figure shows that housing prices rose more in high-amenity regions after 2017. These areas display stronger post-reform increases than lower-amenity regions, consistent with the positive, though marginally significant DiD estimate. While descriptive, the spatial pattern mirrors the econometric finding of slightly larger price growth in high-amenity regions.

Housing-burden inequality

Column (2) of Table 3 reports the corresponding regression for the share of income spent on housing

$$\hat{\theta} = 0.198 \text{ (s.e. } 0.093, p = 0.047\text{)}.$$

High-amenity regions experienced a 0.2 percentage-point post-2017 increase in housing-cost burdens relative to low amenity regions. The magnitude is small in absolute terms but meaningful relative to baseline regional variation (~1.3 p.p. SD; Appendix B). This effect is statistically significant at the 5% level, suggesting that modest demand pressures can translate into measurable affordability differences.

Continuous Event-Study Analysis

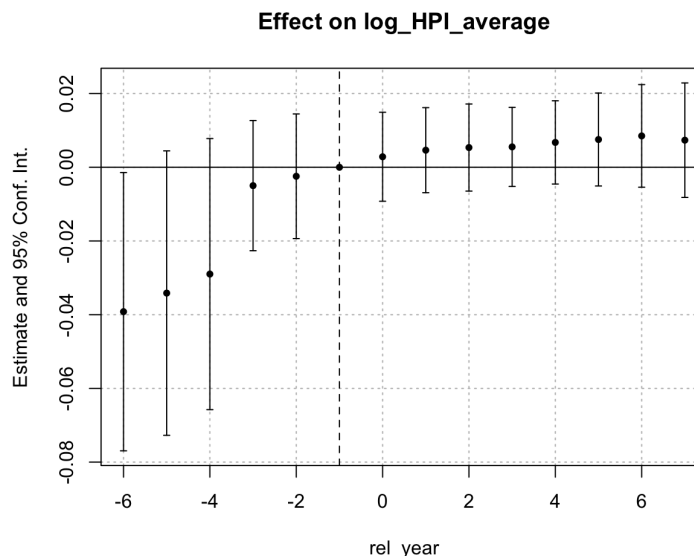


Figure 5. Continuous Event-Study Estimates for Housing Prices. Coefficient estimates from a continuous event-study regression interacting the *AmenityIndex* with relative-year indicators around 2017 (reference year = -1). Error bars show 95% confidence intervals. Pre-2017 coefficients show no anticipatory trends, and post-treatment effects are small and imprecisely estimated.

Figure 5 presents a descriptive diagnostic of a continuous event study, given treatment intensity is time-invariant. Pre-2017 coefficients are generally negative, but with wide confidence intervals and includes zero, indicating no statistically detectable divergence pre-reform. Post-2017 estimates move towards zero or slightly positive, but remain statistically indistinguishable from zero in most years. The dynamic pattern therefore suggests a weak and noisy adjustment, consistent with the modest magnitude of the policy shock.

Interpretation

Taken together, the results offer mixed support for the hypotheses derived from spatial equilibrium. H1 receives weak evidence: housing prices rose slightly faster in high-amenity regions post-2017. By contrast, H2 is supported: housing-cost burdens increased significantly more in high-amenity regions, consistent with small demand shocks translating into local affordability differences. Visual pre-trend comparisons and the event-study diagnostics support the plausibility of the parallel trends assumption, indicating no evidence of divergent trends before 2017.

Conclusion

This study evaluated the effect of Italy's 2017 non-dom tax regime through the lens of spatial equilibrium. Building on Rosen-Roback models and literature on tax-induced mobility (Colas and Hutchinson 2021), we assessed whether the reform disproportionately affected housing prices and housing-cost burdens in high-amenity regions. The international tax literature shows that such policies can influence mobility among top earners (Kleven et al. 2020), but much less is known about the local general equilibrium consequences.

The evidence indicates modest and heterogeneous effects. Housing prices rose slightly faster in high-amenity regions post-2017, though with marginal significance, while housing cost burdens rose significantly more, indicating mild affordability pressures. Event-study diagnostics reveal no anticipatory patterns, reinforcing the validity of the empirical design.

Important limitations remain, most notably the absence of supply side which are a substantial factor in rent response (Hsieh and Moretti, 2019). A further point would be the limitation of pre-policy data to 2011 limiting trend analysis.

These findings suggest that Italy's non-dom regime was too small in scale to meaningfully shift regional housing prices but did create marginal distributional pressures concentrated in already desirable areas. More broadly, the results highlight that preferential tax regimes can have localised affordability implications, even when aggregate market effects appear limited.

References

- Agenzia delle Entrate, "Quotazioni Immobiliari" (2025) [Data Set - Access through Reserved Area]
<https://www.agenziaentrate.gov.it/portale/schede/fabbricatiterreni/omi/banche-dati/quotazioni-immobiliari>
- Allen, Treb, and Costas Arkolakis. "Economic activity across space: a supply and demand approach." *The Journal of Economic Perspectives* 37, no. 2 (2023): 3-28. doi:10.1257/jep.37.2.3.
- Colas, Mark, and Kevin Hutchinson. "Heterogeneous workers and federal income taxes in a spatial equilibrium." *American Economic Journal: Economic Policy* 13, no. 2 (2021): 100-134.
doi:10.1257/pol.20180529.
- Cun, Wukuang, and M. Hashem Pesaran. "A Spatiotemporal Equilibrium Model of Migration and Housing Interlinkages." *Journal of Housing Economics* 57 (2022). doi:10.1016/j.jhe.2022.101839.
- De Chaisemartin, Clément, and Xavier d'Haultfoeuille. "Two Way Fixed Effects and Difference-in-Differences with Continuous Treatment Intensity." *Journal of Econometrics*, Elsevier, vol. 236(2). (2023). doi: 10.1016/j.jeconom.2023.105480
- Diamond, Rebecca. "The determinants and welfare implications of US workers' diverging location choices by skill: 1980–2000." *American economic review* 106, no. 3 (2016): 479-524.
doi:10.1257/aer.20131706.
- Eurostat. "Nights spent at tourist accommodation establishments (TOUR_OCC_NINAT) [Data Set].
European Commission.
https://ec.europa.eu/eurostat/databrowser/view/tour_occ_ninat/default/table?lang=en
- Fonte, Giuseppe, "Italy to Raise by 50% 'Flat' Tax on Ultra Rich to 300,000 Euros, Officials Say,"
Reuters, October 17, 2025,
<https://www.reuters.com/business/finance/italy-raise-flat-tax-super-rich-300000-euros-officials-say-2025-10-17/>. *Reuters*.
- Glaeser, Edward L, and Joshua D Gottlieb. "The wealth of cities: Agglomeration economies and spatial equilibrium in the United States." *Journal of Economic Literature* 47, no. 4 (2009): 983-1028.
doi:10.1257/jel.47.4.983.

Henley & Partners, "Country Wealth Flows: Wealth Migration 2025," Henley & Partners.
<https://www.henleyglobal.com/publications/henley-private-wealth-migration-report-2025/country-wealth-flows>. Henley & Partners.

Hsieh, Chang-Tai, and Enrico Moretti. "Housing Constraints and Spatial Misallocation." *American Economic Journal. Macroeconomics* 11, no. 2 (2019): 1–39. doi:10.1257/mac.20170388.

ISTAT. "Crimes reported by the police forces to the judicial authority (per 100,000 population), NUTS-2 level" (2025). [Data Set]. *Italian National Institute of Statistics*.
https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1_Z0840JUS_1.0/JUS_CRIMINAL/DCCV_D ELITTIPS/IT1_73_67_DF_DCCV_DELITTIPS_9.1.0

Kleven, Henrik, Camille Landais, and Emmanuel Saez. "Taxation and international migration of superstars: Evidence from the European football market." *American Economic Review* 103, no. 5 (2013): 1892-1924. doi:10.1257/aer.103.5.1892.

Kleven, Henrik, Camille Landais, Mathilde Muñoz, and Stefanie Stantcheva. "Taxation and migration: Evidence and policy implications." *Journal of Economic Perspectives* 34, no. 2 (2020): 119-142. doi:10.1257/jep.34.2.119.

KPMG, "Italy: Non-domiciled Residents — Tax Alert," KPMG, January 11, 2017,
<https://assets.kpmg.com/content/dam/kpmg/xx/pdf/2017/01/tnf-italy-jan11-2017.pdf>. KPMG Assets

Roback, Jennifer. "Wages, rents, and the quality of life." *Journal of Political Economy* 90, no. 6 (1982): 1257-1278. doi:10.1086/261120.

Appendices

Appendix A. Theory Derivation - Spatial Equilibrium Framework

We interpret the 2017 Italian non-dom tax regime through the lens of Rosen-Roback (1982) spatial equilibrium model, in which individuals choose locations to equalise indirect utility:

$$U_r = w_r + A_r - p_r,$$

Here, w_r denotes local wages, A_r represents amenities, and p_r is the housing cost. In spatial equilibrium, regions that offer higher amenities must exhibit correspondingly higher housing prices, ensuring that utility is equalised across space. A positive shock to amenities ($\Delta A_r > 0$) generates excess demand for housing, causing higher equilibrium housing prices ($\Delta p_r > 0$).

The Italian non-dom regime functions as a fiscal amenity shock: the reform increases the effective attractiveness of Italian regions for internationally mobile, high-income households by substantially reducing their tax burden. Since migrants with high purchasing power are particularly sensitive to lifestyle amenities, the shock disproportionately increases the demand of already desirable regions. These regions should therefore experience the largest post-reform increases in housing demand and prices.